STORMDOM

Global Storm Tracker

Tropical Storm ODALYS Tracker

Hurricane POLO Tracker

Super Typhoon SURIGAE Tracker

Hurricane NOLO Tracker

Tropical Depression FAY Tracker

Tropical Storm RACHEL Tracker

Graphics produced by Climate Prediction Center

See my new article: *How I Met Hurricane Camille*

What Is a Hurricane?

A hurricane (or typhoon, or severe tropical cyclone), the strongest storm on Earth, is a cyclonic (rotary) storm that derives its energy from cloud formation and rainfall, unlike frontal cyclones that derive their power from a temperature contrast.

A hurricane begins as a tropical depression with a sustained wind speed of less than 39 mph (33 knots, 63 km/h). As the system strengthens, it becomes a tropical storm with winds from 39 to 73 mph (34–63 knots, 63–118 km/h). Tropical storms are named in the Atlantic, East, Central and Northwest Pacific, in the South Indian Ocean, and in the Arabian Sea. When the winds are sustained (based on a one-minute average) at 74 mph (64 knots, 119 km/h), the storm is called a Hurricane (in the Atlantic, East Pacific, Central Pacific or the area between the Dateline and Southeast Pacific (east of 160°E) a **Hurricane**; in the Northwest Pacific (west of the International Dateline) a **Typhoon**; in the Southwest Pacific (west of 160°E) and Southeast Indian Ocean (east of 90°E) a **Severe Tropical Cyclone**; in the North Indian Ocean a **Severe Cyclonic Storm**; and in the Southwest Indian Ocean (west of 90°E) a **Tropical Cyclone**.

The Saffir-Simpson Hurricane Scale

Category 1 – 64-82 knots (74-95 mph, 119-153 km/h). Damage is limited to foliage, signpost, unanchored boats and mobile homes. There is no damage to buildings. Coastal flooding. The main threat to life and property comes from heavy rain.

Category 2 – 83-95 knots (96-110 mph, 154-177 km/h). Roof damage to buildings. Doors and windows damaged. Mobile homes severely damaged. Piers destroyed. Trees torn down. Extensive coastal flooding.

Category 3 – 96-112 knots (111-129 mph, 178-208 km/h). Major Hurricane. Structural damage to some buildings. Mobile homes are completely destroyed. Roof damage is common. Storm surge begins to cause significant damage in beaches and harbors, with small buildings destroyed.

Category 4 – 113-136 knots (130-156 mph, 209-251 km/h). Structural failure of some buildings. Complete roof failures on many buildings. Extensive storm surge damage and flooding. Major erosion of beaches. Severe damage to lower floors of buildings near the shoreline. Hurricane force winds extend well inland.

Category 5 – 137+ knots (157+ mph, 252+ km/h). Complete roof failure on most buildings. Many buildings destroyed, or structurally damaged beyond repair. Catastrophic storm surge damage. In the Northwest Pacific, a typhoon that reaches 150 mph (241 km/h) is called a Super Typhoon.

SAFFIR-SIMPSON SCALE				
Category	Knots	MPH	KM/H	Damage
1	64-82	74-95	119-153	Minimal
2	83-95	96-110	154-177	Moderate
3	96-112	111-129	178-208	Extensive
4	113-136	130-156	209-251	Extreme
Super Typhoon	150+	173+	241+	Catastrophic
5	137+	157+	252+	Catastrophic

Storm Surge

Historically, storm surge is the primary killer in hurricanes. The exact storm surge in any given area will be determined by how quickly the water depth increases offshore. In deep-water environments, such as the Hawaiian islands, storm surge will be enhanced by the rapidly increasing wind speeds in the wind-driven waves associated with the storm. In areas where the ocean bottom has a gradual incline (left-front in the Southern Hemisphere) of the eyewall landfall, where on-shore winds are the strongest, and at the leading edge of the eyewall. Contrary to a popular belief, the storm surge is entirely wind-driven and has little to do with the pressure of the eye. Another factor is the severity of the storm surge is tide. Obviously, an 18-foot storm surge at high tide is that much worse than an 18-foot surge at low tide.

Tropical Cyclone Formation

By Jonathon Edwards

Tropical Cyclone Genesis is the technical term for the process of storm formation that leads ultimately to what are called hurricanes, typhoons, or tropical cyclones in various parts of the world.

This occurs when, in the Northern Hemisphere, the Intertropical Convergence Zone, or ITCZ, shifts northward out of the doldrums and atmospheric conditions become favorable for tropical cyclone formation about after the month of May.

A series of low-pressure ripples develops within the ITCZ. These are known as tropical waves and progress from east to west. In the late season, they typically shift their movement toward the west-northwest, or even northwest, after crossing 45° or 50° W longitude.

These tropical waves, ideally embedded in the deep-layer easterly flow, contain a northeast wind shift. This is typically referred to as a trade-wind circulation, which flows off equal distances from the subtropical ridge (the high-pressure ridge over the north) and the developing low-pressure system. The divergence that results ahead of the convergence zone gives us a subtropically wind as the axis of the tropical wave approaches. Gusts up to 25 mph may occur. Sometimes there can be gusts to tropical storm force in stronger waves. There will be next to no weather associated with these waves, and they may pass virtually unnoticed. More typically, there are bands of disturbed weather ending the axis of the wave.

Tropical cyclones have the low-level circulation and the upper-level circulation (outflow), whose formation was discussed above. There is also a mid-level circulation. The influence of vertical wind shear on the tropical cyclone is the single most important factor in the development of a tropical cyclone that is passing over land. The lower-level circulation can be severely disrupted, or even dissipated, by interaction with land, especially mountainous terrain. If the mid-level circulation remains intact, the upper-level circulation may regenerate a surface center, assist in the surface, it must also evacuate this inbound flow aloft. This occurs in the upper levels of the atmosphere, where high pressure
facilitates the development of the cyclone by evacuating the flow from the lower levels of the cyclone. Every powerful hurricane has an equally powerful high pressure system over it. At the surface, the air spirals inward in a counter-clockwise direction, rises in the developing center, and spirals out at the top in a clockwise direction. In the Southern Hemisphere, it is reversed: clockwise inward, counterclockwise outbound.
If the upper-level high pressure system does not remain over the cyclone, it means there is shear ahead. This is a strong jet or air that is moving rapidly from west to east toward the top of the deep convection. This has the effect of breaking down the whole mechanism. This is known as vertical shear. Vertical shear usually comes from a westerly or southerly direction, and can occur if the cyclone is located in the southern portion of a cold trough or upper-level low pressure system. Another factor that can interfere with the development of a tropical cyclone is subsidence. Subsidence is the sinking of air. This happens on the edge of an upper-level high-pressure system. Subsiding air has the effect of suppressing thunderstorm formation. This also is why a tropical storm that enters the coastal waters of established hurricanes has a very difficult time—the cyclone is on the edge of the hurricane's upper-level outflow, and may have to contend with both subsidence and shear. The effect of shear on a cyclone can range from a failure to develop further to catastrophic collapse of the tropical cyclone's upper structure.

Moving over cool water. Similar to upwelling, when a tropical cyclone moves over cool water (below 77° Fahrenheit/25° Celsius), it begins to weaken. Eventually this causes dissipation, particularly in the East Pacific. In the Atlantic, if the storm is caught in the mid-latitude westerlies and begins to recurve (move eastward), it generally becomes extratropical since it is moving over cooler water. This can be a dangerous storm, however, as it can undergo transformation into a powerful, cold-core, low-pressure system. Though devoid of tropical characteristics, such systems can produce hurricane-force winds, heavy rain, and flooding. Hurricane Hazel in 1954 was such a storm, and eventually became caught in the westerlies and moved northeastward through the Great Lakes and into Canada, where it produced severe flooding in Toronto. The same thing happens when a hurricane passes north of the Azores and hits the British Isles as a Force 8 or stronger gale, having maintained a recognizable inner core.

Exposure to super-level westerlies. Frequently, the ridge of high pressure that keeps the hurricane heading toward the west erodes. In the Atlantic Ocean, the result is strong upper-level westerlies blowing down and impinging on the northern edge of the storm's upper-level support structure. Outward outflow is restricted. A hurricane can still maintain itself when outflow is restricted in one quadrant. If outflow becomes too restricted, resulting shear can disrupt the inner core.

After landfall. Tropical cyclones cannot survive over land, once their access to warm SST is removed. A powerful hurricane, such as Hugo, which hit Charleston, S.C., in 1989 can pose life-threatening problems even after the storm has been reduced to below tropical storm strength. Indeed, it can dump a huge amount of rain—sometimes measured in feet. The storm may evolve into a frontal cyclone that continues to cause widespread damage. The most deadly example of this is Hurricane Camille in 1969—the strongest hurricane ever to make landfall on the continental United States with winds sustained at 190 mph and gusts over exceeding 200 mph—which moved by the Mississippi Valley and eventually merged with a cold front, producing torrential rainfall and flooding in Virginia. Tropical storm Agnes in 1972 produced similar flooding in the Northeast. Hurricane Diane, which developed into a "super cyclone" just north of Australia. The northerly (equatorward) inflow over the Arafura Sea fueled this storm.

What are the factors that contribute to the decay of a tropical cyclone? They are Upwelling, Entraining dry air, Moving over cool waters, Exposure to upper-level westerlies, Interaction with land, and Eyewall replacement cycles.

Upwelling. When a hurricane stalls, its movement is has fallen below 5mph, or its movement is erratic over a small area, the wave action caused by the strong surface winds churns the ocean surface and produces upwelling. This has the effect of cooling the temperature of the sea surface by as much as 5°F within the next 300 miles across the storm's path. Upwelling is especially a problem in the area just ahead of the storm's forward motion. Though the storm will eventually begin to move again, it generally becomes extratropical when it has traversed its own cold wake. Though devoid of tropical characteristics, such systems can produce hurricane-force winds, heavy rain, and flooding. Hurricane Hazel in 1954 was such a storm, and eventually became caught in the westerlies and moved northeastward through the Great Lakes and into Canada, where it produced severe flooding in Toronto. The same thing happens when a hurricane passes north of the Azores and hits the British Isles as a Force 8 or stronger gale, having maintained a recognizable inner core.

Entraining dry air. Sometimes, during the peak season, when tropical cyclones approach continental land masses, they may entrain dry air as they move northward. Storms with frontal troughs that cut off, dry air behind them. The dry air kills off the convection that drive the hurricane's engine. If the dry air entrains deeply enough, the result can be dissipation.

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